Database Systems

Assignment 03 – Fall 2024

Instructions:

1. Please submit a pdf file on LMS with the name format: DB-A03-YOUR_REGISTRATION_NO-Fall2024.pdf (e.g. DB-A03-Numl-F23-12345).
2. Note: Only the placeholders in red will be replaced with your respective ones.
3. Make sure your complete names and roll numbers are clearly mentioned on the title page of the deliverable that you submit.
4. If your submission is late or doesn't follow the naming scheme/instructions, it will not be graded.
5. Due Date: 11:59 pm - December 21, 2024.
6. This is an individual hand-written assignment. Submit a scanned copy.

You are tasked with designing a database for a university library system. The library keeps records of the books issued to students and the respective librarians who facilitate the issuance. Below is the raw, un-normalized data:

StudentID	StudentName	Course	BooksIssued	Author	LibrarianName	LibrarianContact
201	Alice	CS101	Database Systems, Programming in C++	Elmasri, Balagurusamy	John	555-1234
202	Bob	CS102	Operating Systems, Data Structures	Silberschatz, Tanenbaum	Emily	555-5678
201	Alice	CS101	Machine Learning	Mitchell	John	555-1234
203	Carol	CS101	Database Systems	Elmasri	John	555-1234

Tasks

Part 1: Identify Issues and Convert to 1NF

1. Identify the issues in the table that violate the principles of 1NF.
2. Transform the table into 1NF, ensuring atomicity and removing any multi-valued attributes.

Part 2: Decompose into 2NF-Compliant Tables

1. Analyze the 1NF table to identify partial dependencies.
2. Create separate tables to remove partial dependencies, ensuring each table is in 2NF.

Part 3: Normalize to 3NF

1. Examine the 2NF tables for transitive dependencies.
2. Decompose the 2NF tables into 3NF by removing transitive dependencies, ensuring each table only contains attributes dependent on the primary key.

Expected Outputs

- **1NF:** Each cell must have atomic values (no lists or sets).
- **2NF:** The tables should not have partial dependencies.
- **3NF:** The tables should not have transitive dependencies.